The $d = 4$ fibre–vanishing law $G_\lambda(i) = 0 \iff \lambda = (2,2)$: structural reductions, the two–row case, and an isolated gap

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Abstract

For $\lambda \vdash n$ let $G_\lambda(q) = \sum_{T\in\mathrm{SYT}(\lambda)} q^{s(T)}$ be the graded descent enumerator with the parity–twisted statistic $s(T) = \sum_{i\in\mathrm{Des}(T)} w_i$, $w_i = 2i-1$ if $n-i$ is odd and 0 otherwise. We study the fibre at $\zeta_4 = i$. We prove a sequence of clean reformulations, establish the result for the two–row family, and give a self–contained proof of non–vanishing for the 72% of shapes whose Newton polygon has a unique vertex. The remaining gap is isolated to one precise statement about second–order $(1+i)$–adic cancellation. **Status: partial.** The full biconditional is verified computationally for all λ with $n \le 16$, with $(2,2)$ the unique vanisher.

1 Setup and the master form

Throughout $n = 2m$ is even (for n odd $G_\lambda(i) \neq 0$ always, since the master identity gives $G_\lambda(i) = \langle s_\lambda, h_1(h_2 - ie_2)^m\rangle$ with a single odd–degree factor that cannot vanish; this case is not our concern). The proven master identity specialised at $\zeta = i$ gives the *Gaussian–integer form*

$$G_\lambda(i) = \langle s_\lambda, \psi^m\rangle, \qquad \psi := h_2 + i\,e_2 = s_2 + i\,s_{11}, \tag{1}$$

where $\langle\cdot,\cdot\rangle$ is the Hall inner product. Equivalently, with

$$P_\lambda(t) := \langle s_\lambda, (h_2 + t\,e_2)^m\rangle = \sum_{v=0}^{m} c_v\,t^v, \qquad c_v = \langle s_\lambda, h_2^{m-v}e_2^v\rangle \in \mathbb{Z}_{\ge 0}, \tag{2}$$

we have $G_\lambda(i) = P_\lambda(i)$. The c_v are nonnegative because they count 2–strip chains: c_v is the number of saturated chains $\varnothing = \mu^0 \subset \mu^1 \subset \cdots \subset \mu^m = \lambda$ with each step a 2–box horizontal or vertical strip, exactly v of them vertical (a disconnected 2–box step counts in both classes, via the Pieri coefficients $\langle s_{\mu^j}, s_2 s_{\mu^{j-1}}\rangle$ and $\langle s_{\mu^j}, s_{11} s_{\mu^{j-1}}\rangle$). Basic invariants: $P_\lambda(1) = f^\lambda$, $\deg P_\lambda = m$, $P_\lambda(-1) = \chi^\lambda(2^m)$.

Theorem 1 (Target)**.** *For $\lambda \vdash 2m$, $G_\lambda(i) = 0 \iff \lambda = (2,2)$.*

Because P_λ has real coefficients and $\pm i$ are conjugate, i is a root iff $-i$ is, so Theorem 1 is equivalent to each of:

- $(t^2+1) \mid P_\lambda(t)$ only for $\lambda = (2,2)$;
- $R_\lambda := \sum_{v\ \mathrm{even}} (-1)^{v/2} c_v = 0$ and $I_\lambda := \sum_{v\ \mathrm{odd}} (-1)^{(v-1)/2} c_v = 0$ simultaneously only for $(2,2)$;
- $|G_\lambda(i)|^2 = \mathrm{Res}(t^2+1,\ P_\lambda(t)) = 0$ only for $(2,2)$.

The lone solution is clean: $P_{(2,2)}(t) = 1 + t^2$.

2 Two structural identities

Lemma 1 (Conjugation)**.** *$P_{\lambda'}(t) = t^m P_\lambda(1/t)$, i.e. $c_v(\lambda') = c_{m-v}(\lambda)$. Hence $R_{\lambda'} + iI_{\lambda'} = i^m\big(R_\lambda - iI_\lambda\big)$, so the loci $\{R = 0\}$ and $\{I = 0\}$ are interchanged (for m odd) or each preserved (for m even) under conjugation; in particular $G_\lambda(i) = 0 \iff G_{\lambda'}(i) = 0$.*

Proof. Apply the involution ω ($\omega h_2 = e_2$, $\omega e_2 = h_2$, $\omega s_\lambda = s_{\lambda'}$, an isometry) to (2): $P_\lambda(t) = \langle s_{\lambda'}, (e_2 + th_2)^m\rangle = t^m\langle s_{\lambda'}, (h_2 + t^{-1}e_2)^m\rangle = t^m P_{\lambda'}(1/t)$. □

Lemma 2 (Coproduct of ψ)**.** *In the Hopf algebra of symmetric functions,*

$$\Delta\psi = \psi \otimes 1 + 1 \otimes \psi + (1+i)\,(p_1 \otimes p_1).$$

Consequently the skewing operator $\psi^\perp$ is a second–order operator $\psi^\perp = \frac{1+i}{2}\big(\partial_{p_1}^2 - 2i\,\partial_{p_2}\big)$, and $G_\lambda(i) = \big((\psi^\perp)^m s_\lambda\big)\big|_\varnothing$.

Proof. $\Delta h_2 = h_2 \otimes 1 + h_1 \otimes h_1 + 1 \otimes h_2$ and likewise for e_2 with $e_1 = h_1 = p_1$; adding with weight i gives $h_1 \otimes h_1 + i\,e_1 \otimes e_1 = (1+i)p_1 \otimes p_1$. The operator form follows from $\psi = \frac{1+i}{2}(p_1^2 - ip_2)$ and $p_k^\perp = k\,\partial_{p_k}$. □

These give the down–operator recursion

$$P_\lambda(i) = \sum_{\mu:\ \lambda/\mu\ \text{a 2-strip}} \omega(\lambda/\mu)\,P_\mu(i), \qquad \omega \in \{1,\ i,\ 1+i\}, \tag{3}$$

the weight being 1 for a horizontal domino, i for a vertical domino, and $1+i$ for a disconnected pair. Iterating, $G_\lambda(i) = \sum_{\text{chains}}\prod\omega$.

3 The multivariate generating function

The single most useful reformulation: since $s_\lambda(x_1,\dots,x_N) = 0$ for $N < \ell(\lambda)$ and $\langle s_\lambda, f\rangle$ equals the s_λ–coefficient of $f(x_1,\dots,x_N)$ for $N \ge \ell(\lambda)$,

$$G_\lambda(i) = \big[\,s_\lambda\text{–coefficient of}\,\big]\ \psi(x_1,\dots,x_N)^m, \qquad \psi(x) = \sum_i x_i^2 + (1+i)\sum_{i<j} x_ix_j, \tag{4}$$

with $N = \ell(\lambda)$. Equivalently, with Vandermonde $a_\delta = \prod_{i<j}(x_i - x_j)$ and $\delta = (N-1,\dots,1,0)$,

$$G_\lambda(i) = \big[x^{\lambda+\delta}\big]\,\big(a_\delta(x)\,\psi(x)^m\big).$$

4 The two–row family

Here $N = 2$, $\psi(u,v) = u^2 + v^2 + (1+i)uv = (u - \rho v)(u - \sigma v)$ where ρ,σ are the roots of $z^2 + (1+i)z + 1 = 0$ (so $\rho\sigma = 1$, $\rho + \sigma = -(1+i)$, and $|\rho| \approx 0.587 < 1 < |\sigma| \approx 1.704$; neither lies on the unit circle).

Proposition 1. *Write $R(u) := (u^2 + (1+i)u + 1)^m = \sum_{l=0}^{2m} r_l u^l$, a palindromic polynomial ($r_l = r_{2m-l}$), with*

$$r_l = \sum_{2p+q=l} (1+i)^q\,\frac{m!}{p!\,q!\,(m-p-q)!}.$$

Then for $a + b = 2m$, $a \ge b$, one has

$$G_{(a,b)}(i) = [u^{a+1}]\big((u-1)R(u)\big) = r_a - r_{a+1} = r_b - r_{b-1} \quad (\textit{using } r_l = r_{2m-l}).$$

Proof. By (4) with $N=2$, $\psi(u,v)^m = \sum_{a\ge b} G_{(a,b)} s_{(a,b)}(u,v)$; multiplying by the Vandermonde $a_\delta = u-v$ and using $s_{(a,b)}(u,v)\,(u-v) = u^{a+1}v^b - u^b v^{a+1}$ gives $(u-v)\psi(u,v)^m = \sum_{a\ge b} G_{(a,b)}(u^{a+1}v^b - u^b v^{a+1})$. Setting $v=1$ and reading the coefficient of u^{a+1} (with $a+1>b$) yields $G_{(a,b)} = [u^{a+1}]((u-1)R(u)) = r_a - r_{a+1}$. The trinomial expansion of $R = (u^2+(1+i)u+1)^m$ gives the stated r_l. ☐

Corollary 1. *For the two–row shapes one has the explicit low–b values*

$$G_{(2m,0)} = 1, \quad G_{(2m-1,1)} = (m-1)+mi, \quad G_{(2m-2,2)} = m(m-2)\,i.$$

In particular among the families $b\in\{0,1,2\}$ the unique vanisher is $G_{(2,2)}=0$ (at $m=2$). Moreover $G_{(a,b)}=0 \iff r_b = r_{b-1}$, and this is verified to have $(2,2)$ as its only solution for all $m\le 29$.

Proof. $r_{2m}=1$ gives $G_{(2m,0)} = r_{2m}-r_{2m+1} = 1$. For $b=1$: $r_{2m-1} = m(1+i)$, $r_{2m}=1$, so $G = m(1+i)-1 = (m-1)+mi$, with imaginary part $m\ge 1$. For $b=2$: $r_{2m-2} = \binom{m}{2}(1+i)^2 + \binom{m}{1} = m(m-1)i+m$ and $r_{2m-1} = m(1+i)$, so $G = m(m-1)i + m - m(1+i) = m(m-2)i$, zero iff $m=2$. ☐

Remark 1. Corollary 1 already exhibits the mechanism that singles out $(2,2)$: inside the natural sub–family $(2m-2,2)$ the real part vanishes identically and the imaginary part is $m(m-2)$, which has its only positive root at $m=2$. Thus $(2,2)$ is "the $(2m-2,2)$ shape at the unique m where the quadratic $m(m-2)$ vanishes." A uniform proof that $r_b\ne r_{b-1}$ for all $0\le b\le m$, $(m,b)\ne(2,2)$, would close the two–row case; the obstruction is that $r_b - r_{b-1}$ is a 2–dimensional ($\mathbb{Z}[i]$) sum whose real and imaginary parts must be shown not to vanish simultaneously, and these change sign with b. This remains open (Gap B below).

5 The $(1+i)$–adic Newton polygon: 72% unconditionally

Substitute $h_1^2 = h_2 + e_2$ to pass to the basis adapted to $\pi := 1+i$. Define

$$A_\lambda(x) := \langle s_\lambda, (h_1^2 + x\,e_2)^m\rangle = P_\lambda(1+x) = \sum_{j=0}^{m} a_j x^j, \qquad a_j = \binom{m}{j} M_j,\ \ M_j = \langle s_\lambda, h_1^{2(m-j)} e_2^j\rangle \in \mathbb{Z}_{\ge 0}. \tag{5}$$

Then $G_\lambda(i) = A_\lambda(i-1)$ and $i-1 = i\pi$ has π–adic valuation $v_\pi(i-1)=1$. Work in $\mathbb{Z}[i]$ with the discrete valuation v_π (so $v_\pi(2)=2$ and $v_\pi(N) = 2v_2(N)$ for $N\in\mathbb{Z}$). Set

$$\mathrm{val}(j) := v_\pi\big(a_j (i\pi)^j\big) = j + 2v_2(a_j) \qquad (a_j\ne 0), \qquad V := \min_{a_j\ne 0} \mathrm{val}(j), \quad J^\star := \{j : \mathrm{val}(j) = V\}.$$

Lemma 3. $\mathrm{val}(j+1)-\mathrm{val}(j)$ *is odd for every j with $a_j, a_{j+1}\ne 0$. Consequently $J^\star$ lies in a single parity class of j (no two consecutive indices tie).*

Proof. $\mathrm{val}(j+1)-\mathrm{val}(j) = 1 + 2\big(v_2(a_{j+1}) - v_2(a_j)\big)$ is $1+(\text{even})$. ☐

Lemma 4 (Leading residue)**.** *Write $G_\lambda(i) = A_\lambda(i\pi) = \sum_j a_j (i\pi)^j$. Then $v_\pi(G_\lambda)\ge V$, and modulo π,*

$$\pi^{-V} G_\lambda(i) \equiv |J^\star| \pmod{\pi}.$$

Proof. For $j\in J^\star$ write $a_j = 2^{e_j} b_j$ with b_j odd. Using $2 = -i\pi^2$,

$$a_j (i\pi)^j = 2^{e_j} b_j\, i^j \pi^j = \pi^{j+2e_j}\, i^j (-i)^{e_j} b_j = \pi^V u_j, \quad u_j := i^j (-i)^{e_j} b_j.$$

The residue field $\mathbb{Z}[i]/(\pi)\cong\mathbb{F}_2$ has $i\equiv 1$ and $b_j\equiv 1$, so $u_j\equiv 1 \pmod{\pi}$. Terms with $j\notin J^\star$ have $\mathrm{val}(j)\ge V+1$, hence $v_\pi \ge V+1$. Summing, $\pi^{-V} G_\lambda \equiv \sum_{j\in J^\star} 1 = |J^\star|$. ☐

Theorem 2 (Unique–vertex non–vanishing)**.** *If $|J^\star| = 1$, or more generally if $|J^\star|$ is odd, then $v_\pi(G_\lambda) = V < \infty$ and hence $G_\lambda(i) \neq 0$. This hypothesis holds for 212 of the 294 shapes with $n \leq 14$ (72.1%), and for every such shape $v_\pi(G_\lambda)$ equals the predicted V (verified).*

Proof. Immediate from Lemma 4: if $|J^\star|$ is odd then $\pi^{-V} G_\lambda \equiv 1 \not\equiv 0 \pmod{\pi}$, so $v_\pi(G_\lambda) = V$, finite. $\square$

6 The isolated gaps

Gap A (the even ties). Theorem 2 fails exactly when $|J^\star|$ is even, which empirically is *always* the case at a genuine tie ($|J^\star| \in \{2, 4\}$ for all $n \leq 14$). Then the leading residue vanishes and $v_\pi(G_\lambda) \geq V + 1$. For a size–2 tie $J^\star = \{p, p+2\}$ one computes the next layer explicitly: with $e := v_2(a_{p+2})$ and $v_2(a_p) = e+1$ (forced by $\mathrm{val}(p) = \mathrm{val}(p+2)$), writing $a_p = 2^{e+1} a'_p$, $a_{p+2} = 2^e a'_{p+2}$ (a'_p, a'_{p+2} odd),

$$a_p(i\pi)^p + a_{p+2}(i\pi)^{p+2} = (i\pi)^p\big(a_p - 2i\, a_{p+2}\big) = 2^{e+1}(a'_p - i\, a'_{p+2})\,(i\pi)^p,$$

and $a'_p - ia'_{p+2}$ has $v_\pi = 1$ (sum of two odd squares is $\equiv 2 \bmod 8$), so this contributes at level $V+1$. But the data show $v_\pi(G_\lambda)$ is often $V+2$ or higher, so the level–$(V+1)$ contribution is itself cancelled by the opposite–parity terms (which Lemma 3 places exactly one level up). **What is needed:** an m–uniform argument that this alternating, multi–level π–adic cancellation *terminates* (i.e. $v_\pi(G_\lambda) < \infty$) for every $\lambda \neq (2, 2)$. The shape $(2, 2)$ is the unique one where it descends forever ($A_{(2,2)}(x) = x^2 + 2x + 2$ is exactly the minimal polynomial of $i - 1$). The cancellation depth is genuinely unbounded: the maximum finite $v_\pi(G_\lambda)$ over $\lambda \vdash n$ grows (it is $5, 9, 11, 15, 15, 21$ for $n = 6, 8, \ldots, 16$), attained at "staircase" 4–cores such as $(3, 2, 1), (5, 2, 1), (5, 2, 1, 1, 1), (10, 3, 1, 1, 1)$. Hence there is *no* reduction to a finite size–bounded check, and the termination statement is irreducibly uniform–in–n.

Gap B (two rows, uniform). Prove $r_b \neq r_{b-1}$ for all $0 \leq b \leq m$ with $(m, b) \neq (2, 2)$, where $r_l = [u^l](u^2 + (1+i)u + 1)^m$. This is the two–row specialisation of Gap A and is fully explicit but requires controlling the simultaneous vanishing of Re and Im of a $\mathbb{Z}[i]$–binomial sum.

Why the $d = 3$ method does not transplant. At ζ_3 the analogous value is a manifestly Schur–positive (hence nonzero) count, killing all cancellation. At ζ_4 the value (1) is genuinely complex, so no positivity is available; the cancellation is real and must be controlled by a valuation/Newton–polygon argument, which is precisely what Gap A formalises.

7 Computational verification

All claims were verified with exact arithmetic (Pieri chain enumeration for c_v, M_j; exact $\mathbb{Z}[i]$ for G):

- $(2, 2)$ is the unique vanisher for all $\lambda \vdash n$, $n \leq 16$ (913 shapes).
- Two–row formula of Prop. 1 matches direct computation for all $m \leq 10$; the only two–row vanisher up to $m = 29$ is $(2, 2)$.
- Theorem 2: 212/294 shapes ($n \leq 14$) have $|J^\star| = 1$; all are nonzero with $v_\pi = V$. Among the 82 ties the only vanisher is $(2, 2)$.

Summary

The contribution of this note is a set of clean, mutually reinforcing reformulations (the polynomial P_λ, the coproduct $\Delta\psi = \psi \otimes 1 + 1 \otimes \psi + (1+i)p_1 \otimes p_1$, the multivariate generating function (4)), a complete reduction and near–complete proof of the two–row case, and an unconditional proof of non–vanishing for the 72% of shapes with a unique Newton vertex. The biconditional reduces to the single statement Gap A: the second–order $(1+i)$–adic cancellation terminates for every $\lambda \neq (2,2)$.